Ripple

From Headlines to Portfolio Risk: AI/NLP risk engine with Module B stress testing

S&P Global & Crisil
Campus Hackathon 2026

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What Ripple Does

Problem Statment

Build an NLP and risk engine for the S&P Global & Crisil Campus Hackathon 2026 that turns unstructured text into structured risk signals and uses them to drive a downstream module.

Solution

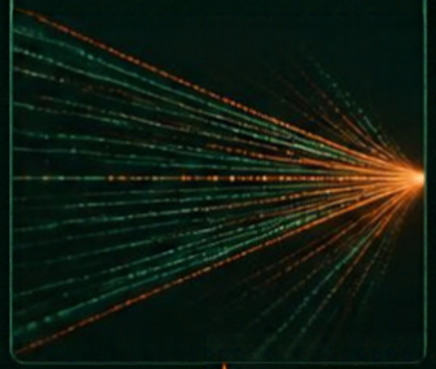
news moves markets before a risk system sees a number, so Ripple scores each headline, raises an alert when an event looks serious, and stress-tests the portfolio to show what it could cost.



Architecture

PUBLIC DATA

GDELT headlines
stock tweets



NLP SCORING

clean and tag text
9 event classes

FinBERT sentiment
impact score 1 to
10

10



JSON OUTPUT

schema-checked
JSONL files
JSON API



MODULE B STRESS ENGINE

synthetic bank book
44 positions
$3.6bn of loans,
bonds, swaps, options



REACT DASHBOARD



hourly as-of scoring,
no look-ahead

shocks equal 2x
the worst week of
historical analogues

IFRS 9
DV01

convexity

Basel III RWA

P&L
attribution

RUNS ON CPU, NO GPU OR API KEYS

Implementation Highlights

01

Ingestion

GDELT plus tweets
hourly replay
plus live feed

02

NLP Scoring

FinBERT with VADER fallback
50 weighted rules into 9 event classes
every number traces to config

03

Stress Engine

calibrated shocks on a 44 position book giving P&L, IFRS 9 and capital

04

Dashboard

world map
risk gauge
replay timeline
signal feed
stress test

Tech Stack Python | Pandas | NumPy | FinBERT | VADER | TypeScript | React | JSON API

Key Results



Replay 10 Feb to 24 Mar 2022

34,298
signals

14
stress tests

Fitch Russia downgrade stress test

Book value down
83.0m
(2.3%)

From **100.0**
to **97.7**

Loss drivers ($m)

-58.2	-42.9	-3.4	-83.0
bond spread widening	equity falls	IFRS 9 provisions	net after rate offset

Ripple vs volume-spike rule, Sep 2022

100%	83%	6	15 h
recall vs 17%	precision	alerts raised vs	median lead

Domain Impact

15 h earlier warning: alerts a median 15 h before an end-of-day batch run.

↓

100% consistent and explainable scoring: every score traces back to a rule or weight.

↓

360° strategic what-if: turns an alert into P&L, provisions and capital.



Who benefits | risk managers | portfolio managers | credit teams | treasury teams

Reproducible and Verified

Frozen before testing
with git tag eval-freeze

95% bootstrap confidence
intervals on every metric

Schema-checked, tested,
deterministic builds

Run it yourself
python run.py starts offline replay
and live GDELT feed
W1, W3 and N1 replay day by day
python -m scriptsverifyFreeze
verifies the freeze

